

AI TRUST DECISION BRIEF

CareerForge Day 5: AI Is Not The Engineer

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GITHUB: <https://github.com/liacodes49/Career-Forge/tree/day-05-ai-is-not-the-engineer>

1. INCIDENT OVERVIEW

An engineering team is responding to a live operational incident. Multiple AI systems have independently analyzed the situation and produced recommendations. The recommendations appear confident and technically detailed. The engineering lead's initial instinct is to follow AI output directly.

This brief evaluates each AI recommendation through the lens of an independent engineer, determining what is trustworthy, what requires verification, what should be modified, and what must be blocked.

2. AI RECOMMENDATIONS RECEIVED

Recommendation A: AI System: Root Cause Diagnosis

Claim: Memory leak in the payments-service pod caused by unbounded connection pool growth. CPU utilization at 94%, OOMKilled events detected in the last 30 minutes.

Confidence: High (AI system cited pod logs + Prometheus alert data)

Recommendation B: AI System: Immediate Fix

Claim: Restart the payments-service pod immediately and increase memory limits from 512Mi to 2Gi. This will resolve the incident within 5 minutes.

Confidence: High

Recommendation C: AI System: Long-term Fix

Claim: Set maxPoolSize=10 in the DB connection configuration and implement exponential backoff on retries. Deploy via automated pipeline without staging validation.

Confidence: Medium

Recommendation D: AI System: Risk Assessment

Claim: This incident has zero blast radius. The payments-service is isolated and a restart carries no downstream risk to user transactions.

Confidence: High

3. TRUST CLASSIFICATION TABLE

Each recommendation has been independently evaluated against available evidence.

Recommendation	AI Claim	Classification	Reason
A. Root Cause	Memory leak from unbounded pool	TRUST (with verification)	Consistent with OOMKilled logs; verify pool metrics first
B. Immediate Fix (restart)	Restart resolves in 5 min	MODIFY	Restart is valid; memory limit increase needs capacity check
B. Memory limit 2Gi	Increase from 512Mi to 2Gi	BLOCK until verified	No capacity baseline provided; could cause node pressure
C. maxPoolSize config	Set maxPoolSize=10	TRUST (with testing)	Reasonable default; must test in staging before prod deploy
C. Skip staging	Deploy without staging	BLOCK	Zero evidence this is safe; staging is mandatory for DB config changes
D. Zero blast radius	Isolated, no downstream risk	BLOCK	Payments service is never truly isolated; requires transaction audit

4. MISSING EVIDENCE LOG

The following information was absent from AI recommendations and must be obtained before acting:

- No transaction in-flight count at time of restart: risk of corrupted payment state unknown
- No node-level memory capacity data: memory limit increase could trigger OOM on host node
- No DB connection audit confirming unbounded pool is the actual root cause vs a symptom
- No staging environment test result for the proposed DB config change
- No rollback plan documented: if restart worsens state, recovery path is undefined
- No downstream dependency map: claims of 'zero blast radius' are unverifiable without this

5. ENGINEERING DECISION

The engineering team should NOT follow any single AI recommendation blindly. The following phased action is recommended:

Step 1: Immediate (0–5 min)

- Pull real-time pod logs and confirm OOMKilled events independently
- Check node memory headroom before any limit changes
- Audit active transaction count before restarting payments-service

Step 2: Controlled Restart (5–10 min)

- Restart the pod only after confirming in-flight transaction count is zero or drained
- Do NOT increase memory limit yet, monitor baseline post-restart first

Step 3: Config Change (via staging, not prod)

- Apply maxPoolSize=10 change in staging; run load test for 10 minutes
- Only promote to prod if staging shows stable pool behavior

Step 4: Post-Incident

- Conduct blast radius audit with downstream teams
- Document actual root cause vs AI's predicted root cause
- Add connection pool monitoring alert as a permanent fixture

6. FINAL VERDICT

AI output provided useful signal but critically insufficient safety evidence. The root cause hypothesis is plausible and worth pursuing. The immediate fix recommendation is partially valid but must be modified. The long-term fix recommendation contains a dangerous deployment shortcut that must be blocked. The blast radius claim is unsupported and must be independently verified before any action is taken.

The AI is not the engineer. The engineer is.